

Exploring Causality in Quantum Computing Systems

Vignesh Thondikulam

REU Program

Project Mentor: Dr. Pooyan Jamshidi

Project Co-Mentor(s): Dr. Mohammad Ali Javidian

July 20, 2026

1 Abstract

Quantum computing workflows involve many configuration choices, such as number of qubits, circuit depth, optimization level, shot count, and simulated noise rate, whose relative influence on quantum program output is rarely made explicit. This project studies how these workflow configurations causally affect quantum program behavior, using controlled experiments and causal analysis. A benchmark suite of five Qiskit circuits (Bell, GHZ, Parameterized, Grover, and Variable Depth) was implemented and executed under systematically varied configurations, producing a dataset of 12,672 runs. A causal graph connecting these configurations, mediating variables, and outcome metrics was constructed and analyzed using DoWhy’s graphical causal model framework, combining arrow strength estimation with counterfactual querying. The analysis found that simulated noise rate is by far the strongest cause of output quality and algorithmic correctness variations: a counterfactual intervention fixing noise rate to a low value improved output quality and Grover search’s success probability by approximately 67.3% and 28%, respectively. Optimization level, by contrast, showed a negligible effect at the circuit scales tested. A Streamlit dashboard was developed to make the dataset, causal graph, and counterfactual findings interactively explorable. These results demonstrate that causal modeling can identify which quantum workflow configurations genuinely matter, and by how much, providing a principled basis for prioritizing noise mitigation over optimization tuning, for the circuits at this scale, in future quantum programming workflows.

2 Introduction

Quantum computing workflows involve a range of choices that affect program outcome behavior in ways that are not always obvious. Circuit size, circuit depth, Qiskit’s transpilation optimization level, number of shots, and simulated noise rate can all influence outcome metrics like output quality, algorithmic correctness, and runtime, yet the importance of these choices and how they interact are not made explicit in typical quantum programming workflows.

This project does not develop a new quantum algorithm. Instead, it studies how workflow configurations affect quantum program behavior using controlled experiments, causal analysis, and counterfactual reasoning, treating the quantum programming pipeline itself as a system to be understood empirically.

The main contributions of this project are:

- a small Qiskit benchmark suite spanning five circuit types (Section 5);
- a dataset of 12,672 quantum program executions under systematically varied configurations (Section 6);
- metrics for output quality and execution behavior (Section 7);
- a causal graph describing relationships among workflow configurations and outcomes (Section 8);
- intervention-style and counterfactual analyses quantifying the relative effect of each configuration (Section 10);
- a lightweight prototype dashboard for exploring the dataset and causal findings (Section 10).

This project was organized around six research questions, concerning how workflow configurations causally influence quantum program output: (1) does increasing circuit depth cause a measurable decrease in output quality under a fixed noise model; (2) does Qiskit’s transpilation optimization level reduce gate count or circuit depth enough to improve output quality; (3) how does the number of shots affect the stability of

the measured output distribution; (4) which has a stronger effect on output quality: circuit depth, two-qubit gate count, or simulated noise rate; (5) can a simple causal graph help explain how quantum workflow configurations influence observed outcomes; and, (6) what would the output quality have been if the same circuit had been run with a lower noise rate or a different optimization level.

3 Background

This section briefly introduces the prerequisite concepts that need to be understood for this project.

3.1 Quantum Programming

A qubit is the basic unit of quantum information, similar to a classical bit but capable of existing in superposition. Quantum circuits are built from gates that manipulate qubits: single-qubit gates (e.g., Hadamard, rotation gates) act on one qubit at a time, while two-qubit gates (e.g., CNOT, CZ gates) create entanglement between qubits. At the end of a circuit, qubits are measured, collapsing their quantum state into a classical bitstring outcome. Because quantum measurement is probabilistic, a circuit is typically executed many times, each execution called a *shot*, to build up a distribution over possible outcomes.

This project uses Qiskit Aer Simulators [4] rather than physical quantum hardware. Simulators can run either as ideal (noiseless) simulators, which reflect the mathematically exact behavior of a circuit, or as noise simulators, which apply a noise model to approximate the errors present in real quantum hardware. This project uses a depolarizing noise model [3], in which gates have a configurable probability of introducing random errors.

Before execution, a quantum circuit is *transpiled*, where it is rewritten into a form compatible with the target backend’s native gate set and, if applicable, optimized for fewer gates or lower depth. Qiskit exposes four transpilation optimization levels (0 – 3), with higher levels applying more aggressive gate reduction and circuit restructuring at the cost of additional compilation time [2].

Readers new to these concepts may find IBM’s introductory quantum computing materials useful [1].

3.2 Workflow Configurations

In this project, a *workflow configuration* refers to any controllable choice made when constructing, compiling, or executing a quantum circuit. The configurations studied here includes the number of qubits, circuit depth, transpilation optimization level, number of shots, and simulated noise rate. These configurations were selected because each represents a distinct stage of the quantum programming pipeline: circuit construction (qubit count, depth), compilation (optimization level), and execution (shots, noise rate), allowing the causal analysis to trace how choices at each stage propagate to outcome metrics.

- **Number of Qubits:** the number of qubits a quantum circuit is initialized with, which determines the size of the circuit’s possible measurement outcomes.
- **Circuit Depth:** the length of the longest sequential chain of gates in a circuit after transpilation.
- **Optimization Level:** an integer (0 – 3) controlling how aggressively Qiskit’s transpiler rewrites a circuit to reduce gate count and depth before execution.
- **Number of Shots:** the number of times an identical circuit is executed and measured to build up a distribution over possible outcomes.

- **Noise Rate:** the probability that a gate or qubit introduces error during execution, modeled here using a depolarizing noise channel. On real hardware, such errors arise from physical factors like temperature and magnetic interference.

These concepts, such as qubits, quantum gates, and measurement, form the standard foundation of quantum computation, as described by Nielsen and Chuang [6].

3.3 Causal Analysis

Causal analysis studies how changes in one variable affect another, going beyond simple correlation to model the underlying mechanisms that generate observed data. A *causal graph* is a directed acyclic graph (DAG) in which nodes represent variables and edges represent assumed direct causal relationships. Causal graphs make modeling assumptions explicit and allow two related but distinct types of queries:

- **Interventions:** estimating the effect of actively setting a variable to a specific value, independent of its normal causes (e.g., asking how output quality would change if noise rate were fixed to a particular value across all runs).
- **Counterfactuals:** estimating what the outcome would have been for a specific observed instance, had one of its causes been different, while holding all other mechanisms fixed as they were actually realized.

This project uses DoWhy’s graphical causal model (gcm) framework [5] in Python (see Code Repository in Appendix B) to perform both types of queries: arrow strength estimation to rank the relative influence of causal variables, and counterfactual sampling to estimate outcomes under hypothetical configuration changes, as detailed in Section 9.

4 Project Goals

Building on the motivation described above, this project pursued the following concrete goals:

1. Implement a small benchmark suite of quantum circuits.
2. Run circuits under systematically varied configurations.
3. Collect structured execution data.
4. Compute quality and performance metrics.
5. Build causal graphs for the workflow.
6. Analyze how configuration changes affect outcomes.
7. Develop a lightweight tool for running and inspecting experiments.

5 Benchmark Suite

Five benchmark circuits were implemented to study causal relationships between workflow configurations and output quality, each chosen to isolate and understand different structural properties relevant to noise sensitivity and transpilation behavior.

5.1 Bell State Circuit

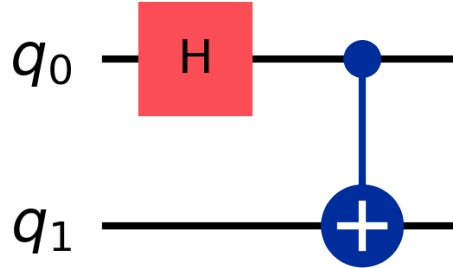


Figure 1: Bell state circuit: a Hadamard gate followed by a CNOT, producing a maximally entangled state, fixed at 2 qubits. Serves as the baseline entanglement benchmark; shot count, noise rate, and optimization level are configurable. Evaluated using TVD and runtime.

5.2 GHZ State Circuit

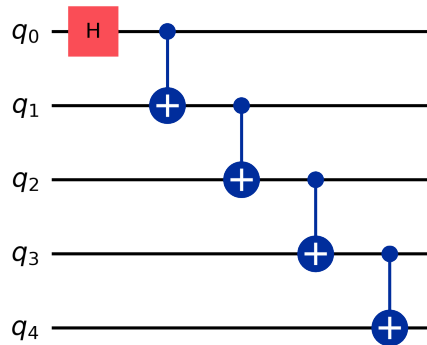


Figure 2: GHZ state circuit (5-qubit configuration as shown): a Hadamard gate on the first qubit followed by a chain of CNOTs, extending entanglement across all qubits. Tests entanglement scaling as qubit count increases. Qubit count, shot count, noise rate, and optimization level are configurable. Evaluated using TVD, depth, 2-qubit gate count, and runtime.

5.3 Parameterized Circuit

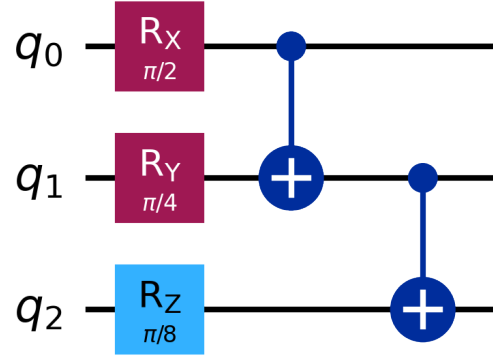


Figure 3: Parameterized circuit (3-qubit configuration as shown), using single-qubit rotation gates (R_X , R_Y , R_Z) followed by entangling CNOTs. Qubit count, shot count, noise rate, and optimization level are configurable. Evaluated using TVD, depth, 2-qubit gate count, and runtime.

5.4 Grover Search

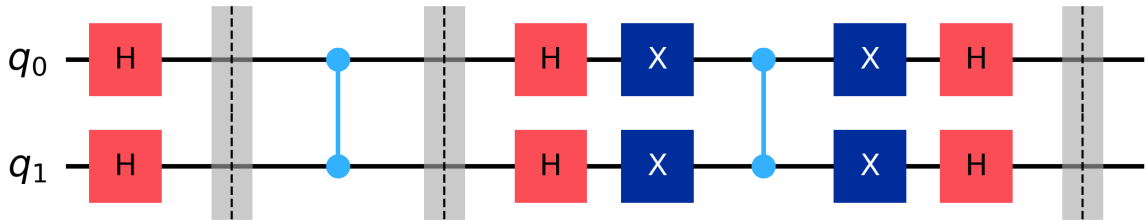


Figure 4: Grover search circuit (2-qubit configuration as shown). Grover's algorithm searches an unsorted space to find a marked target state in $O(\sqrt{N})$ time, a quadratic speedup over the $O(N)$ time required classically. This benchmark tests algorithm correctness under noise. Qubit count, target state, shot count, noise rate, and optimization level are configurable. Evaluated using TVD, success probability, depth, 2-qubit gate count, and runtime.

5.5 Variable Depth Circuit

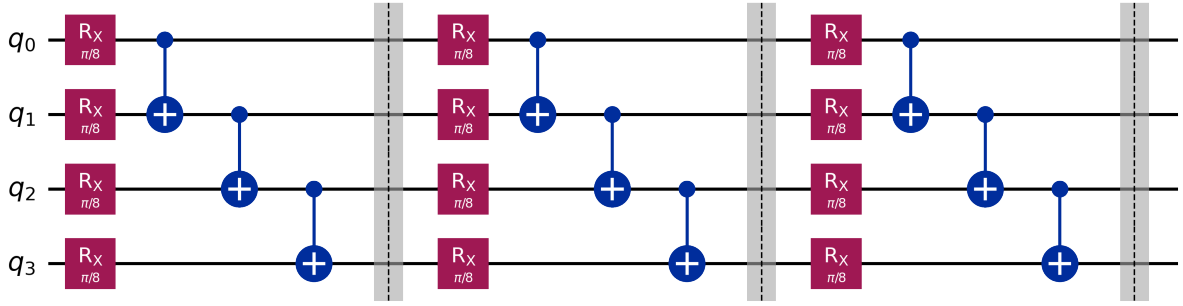


Figure 5: Variable depth circuit (4-qubit configuration as shown), with repeated layers of RX rotations and CNOT chains. Tests depth-noise interaction directly. Qubit count, depth, shot count, noise rate, and optimization level are configurable. Evaluated using TVD, depth, 2-qubit gate count, and runtime.

Table 1: Benchmark Circuits

Circuit	Purpose	Configurations	Metrics
Bell State	Baseline entanglement benchmark	shots, noise rate, optimization level	TVD, runtime
GHZ State	Multi-qubit entanglement scaling	qubits (2 – 5), shots, noise rate, optimization level	TVD, depth, count_2q, runtime
Parameterized	Rotation gate sensitivity	qubits (2 – 5), axis/angle pairs, shots, noise rate, optimization level	TVD, depth, count_2q, runtime
Grover	Algorithm correctness under noise	qubits (2 – 4), target string, shots, noise rate, optimization level	TVD, success probability, depth, count_2q, runtime
Variable Depth	Depth and noise interaction study	qubits (2 – 5), depth (1 – 15), shots, noise rate, optimization level	TVD, depth, count_2q, runtime

6 Experimental Design

Each benchmark circuit was executed under systematically varied workflow configurations to generate the dataset used for causal analysis. The experiment pipeline followed a consistent sequence for every run:

1. Select a benchmark circuit (Bell, GHZ, Parameterized, Grover, or Variable Depth).
2. Select configuration values (qubit count, depth, optimization level, shot count, noise rate).
3. Build the circuit in Qiskit and transpile it using the selected optimization level.
4. Execute the transpiled circuit on a Qiskit AerSimulator configured with the selected noise rate.
5. Collect measurement results from the simulator.

6. Compute outcome metrics (TVD, runtime, depth, gate counts, and success probability where applicable).
7. Store the configuration values and computed metrics (total 12,672) as a row in the structured results dataset (`experiment_results.csv`).

Table 2: Configuration Space

Configuration	Values Used	Purpose
Number of qubits	2, 3, 4, 5	Study how circuit scale affects output quality and runtime
Circuit depth*	1, 2, 3, 5, 7, 10, 15	Study whether deeper circuits accumulate more error under noise
Optimization level	0, 1, 2, 3	Study how transpilation changes circuit depth, gate count, and outcome metrics
Number of shots	128, 256, 512, 1024, 2048, 4096	Study how sampling affects the stability of measurement outcomes
Noise rate	0.0, 0.001, 0.005, 0.01, 0.03, 0.05	Study how simulated depolarizing noise changes program behavior
Grover target [†]	'11' (n=2), '101' (n=3), '1001' (n=4)	Define the search target for Grover's algorithm at each qubit count

*Modeled as a mediator variable (caused by optimization level and circuit structure), therefore, excluded from the causal graph.

[†]Circuit-specific parameter for the Grover benchmark; excluded from the causal graph as it does not generalize across circuits.

It is important to note that the total number of executions (12,672) is smaller than a full factorial design over all listed configuration values would imply. This is because not all parameters are meaningful for every circuit type: circuit depth, in particular, is only a configurable, systematically varied parameter for the Variable Depth circuit, since the other four benchmark circuits (Bell State, GHZ State, Parameterized, and Grover) have a fixed, algorithm-defined depth that does not change across configurations. As a result, the circuit depth values in the configuration table (1, 2, 3, 5, 7, 10, 15) were only swept for the Variable Depth circuit, while the remaining circuits were run at their single native depth across the other varied parameters (number of qubits, optimization level, number of shots, and noise rate). This design choice avoids generating invalid or redundant combinations that would not correspond to a meaningful distinct execution.

7 Metrics

The following metrics were computed for each experimental run to evaluate output quality and execution behavior:

- **Success probability:** probability of observing the expected target output. Applicable only to the Grover Search circuit.
- **Total Variation Distance (TVD):** the difference between the observed and ideal output distributions, used as the primary measure of output quality across all five circuits.
- **Transpiled depth:** circuit depth after transpilation at the selected optimization level.
- **Two-qubit gate count (count_2q):** the number of two-qubit gates after transpilation; used as a good indicator of noise since two-qubit gates typically carry higher error rates than single-qubit gates.
- **Runtime:** the time required to transpile and execute the circuit.

8 Causal Model

The quantum workflow was represented as a directed acyclic graph (DAG) connecting configuration variables, mediating variables, and outcome metrics. The graph consists of 10 nodes and 18 edges, distinguishing three types of variables.

- **Causal variables (roots):** shots, noise_rate, optimization_level, num_qubits, circuit_name
- **Mediators:** depth, count_2q
- **Outcomes:** tvd, runtime, success_probability

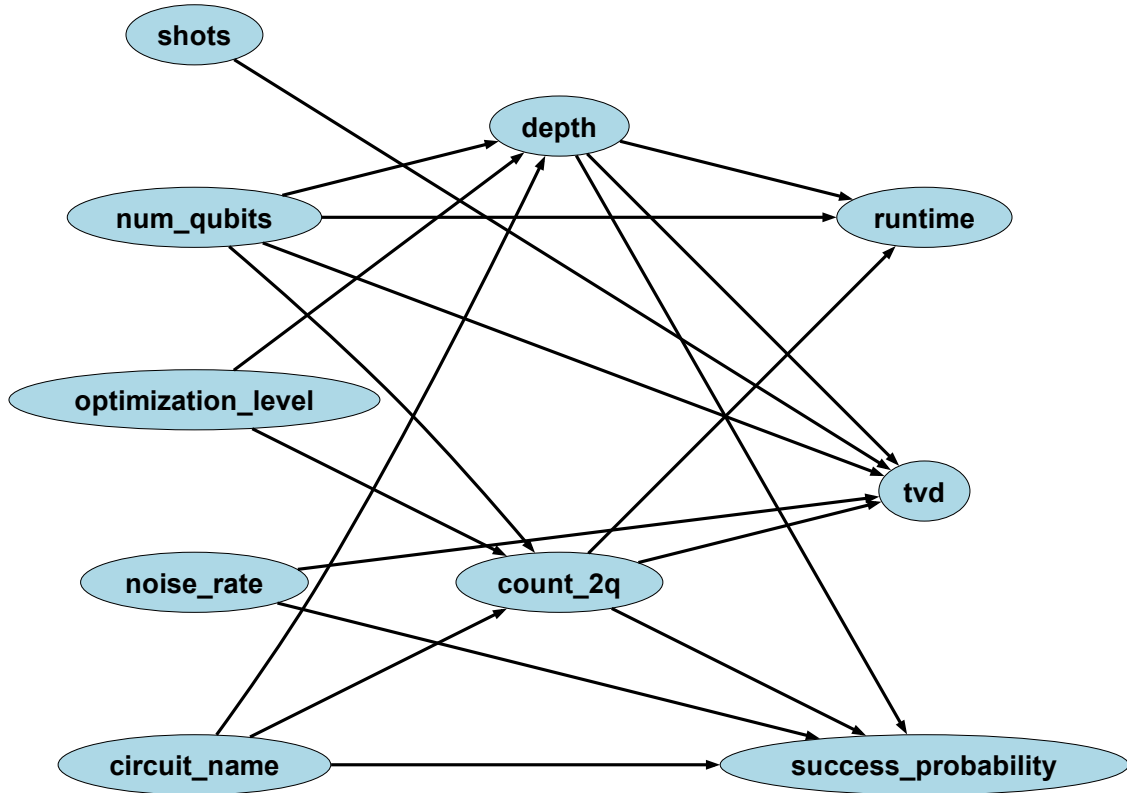


Figure 6: Causal graph for quantum workflow configurations and outcomes.

Several modeling decisions shaped the final graph structure:

- `circuit_name` was retained as a causal variable, since circuit structure directly affects which mediating paths are active.
- `depth_config` was not considered as a causal variable, as it does not apply uniformly across all five benchmark circuits.
- A direct edge from `num_qubits` to `tvd` was confirmed empirically, instead of assuming that its effect is fully mediated through depth or gate count variables.
- The edge from `optimization_level` to `count_2q` was retained as a real but largely dormant mechanism - present in the graph structure but weak in practice.

The causal graph structure was specified using a combination of domain knowledge of the experimental pipeline (e.g., the known dependence of circuit depth and gate count on qubit count, circuit type, and transpilation) and manual inspection of the collected data, drawn from the `experiment_results.csv` file containing 12,672 rows across five benchmark circuits and systematically varied configurations. The resulting structure was subsequently fit and validated within DoWhy’s graphical causal model (`gcm`) framework, which was used to assign causal mechanisms to each node and confirm that the specified graph produced coherent, well-behaved arrow-strength and counterfactual estimates consistent with expected physical relationships (e.g., noise rate showing a strong direct effect on output quality).

9 Causal and Counterfactual Analysis

Using the causal graph (from Section 8) as the basis for analysis, intervention-style and counterfactual queries were performed using DoWhy’s graphical causal model framework (`gcm`). Arrow strength estimation (`gcm.arrow_strength`) was used to rank the relative influence of causal variables on TVD; counterfactual sampling was used to estimate outcomes under hypothetical configuration changes.

9.1 Relative Strength of Causal Variables

Ranking causal variables by their estimated arrow strength on TVD showed a clear ordering: `noise_rate` had the strongest effect, followed by `depth`, `count_2q`, `num_qubits`, and `shots`, which had the weakest effect among the five. The `optimization_level`’s effect on TVD was found to be negligible, consistent with its role as a dormant mechanism identified during causal graph construction (see Section 8).

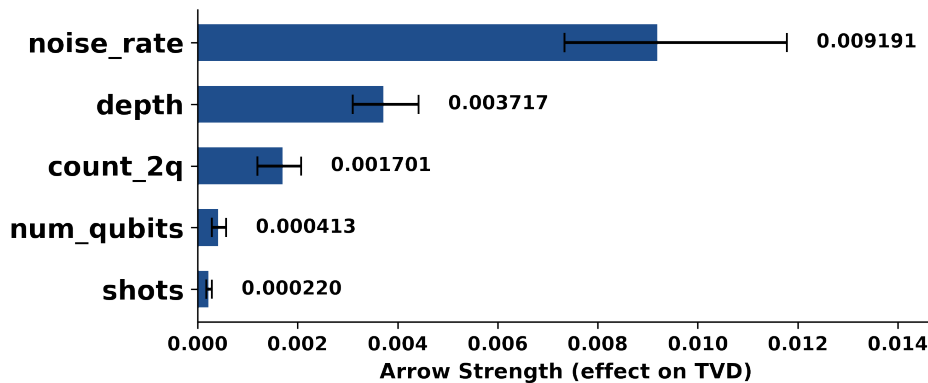


Figure 7: Estimated Causal Influence on TVD (larger = stronger; 95% CI shown).

9.2 Effect of Circuit Depth Under Fixed Noise Levels

To directly address whether increasing circuit depth causes a measurable decrease in output quality under a fixed noise model, TVD was examined as a function of circuit depth at three fixed noise levels (low, medium, high) for the Variable Depth benchmark (Figure 8). At every noise level tested, TVD increased as circuit depth increased, with the effect most pronounced under high noise, where TVD rose from approximately 0.16 at depth 1 to approximately 0.38 at depth 15. This confirms that deeper circuits accumulate more noise-induced error before measurement.

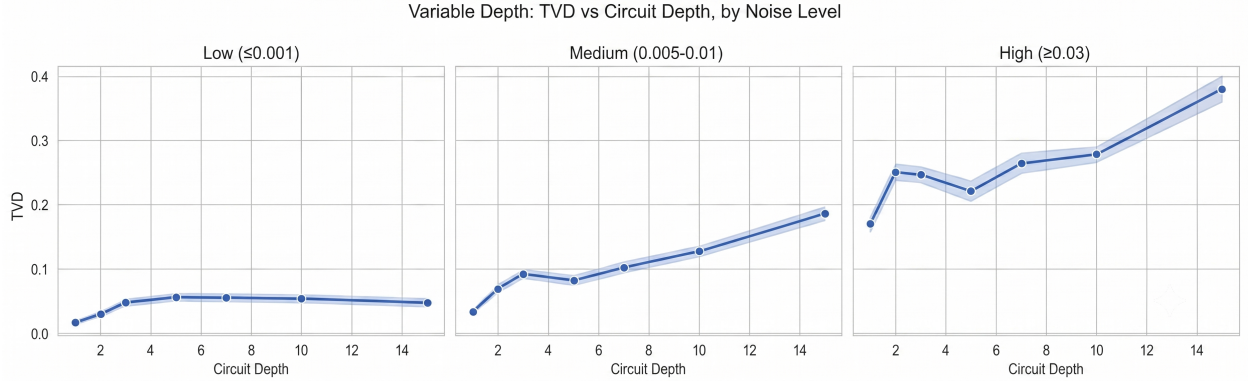


Figure 8: TVD as a function of circuit depth, shown separately for low, medium, and high noise levels on the Variable Depth benchmark circuit.

9.3 Effect of Circuit Depth and Optimization Level

Despite depth ranking as the second-strongest cause of TVD, higher optimization levels did not meaningfully reduce TVD in practice. A counterfactual query fixing `optimization_level` to 3 changed mean TVD (averaged across all five benchmark circuits) from an observed value of 0.10445 to a counterfactual value of 0.10331, a reduction that is around 1% (see Table 3 in Appendix A), indicating that while optimization level can influence depth and gate count, its effect on output quality is weak relative to other variables in the graph. This result reflects the relatively shallow, small-qubit circuits used in this study; optimization level's effect on output quality may be more pronounced for larger, deeper circuits where transpilation has more opportunity to reduce gate count and error accumulation.

9.4 Effect on Noise Rate

Noise rate was confirmed as the dominant driver of output quality degradation. A counterfactual query fixing `noise_rate` to 0.001 changed mean TVD (averaged across all five benchmark circuits) from an observed value of 0.10445 to a counterfactual value of 0.03412, which is a reduction of approximately 67.3%, the largest effect observed among all tested interventions (see Table 3 in Appendix A).

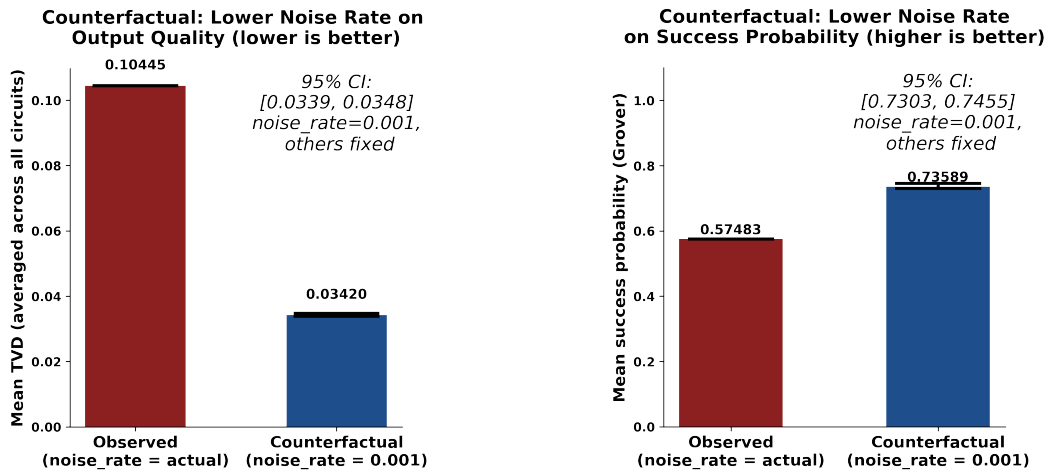


Figure 9: Counterfactual analysis of setting a low noise rate condition, showing mean TVD averaged across all five benchmark circuits compared alongside the Grover-specific success probability performance.

9.5 Effect of Shot Count on Sampling Stability

Shot count ranked weakest among the five causal variables in its effect on TVD (see Figure 9.1), indicating that increasing shots produces comparatively small improvements in output quality, with those improvements exhibiting diminishing returns as shot count grows. This is consistent with shot count primarily affecting measurement sampling stability rather than the underlying error introduced by noise or circuit depth (see Figure 11 in Appendix A).

9.6 Counterfactual Effect on Grover Success Probability

Under the same counterfactual condition (noise_rate fixed to 0.001), the Grover Search benchmark’s success probability changed from an observed value of 0.57483 to a counterfactual value of 0.73580, which is an improvement of approximately 28% (see Table 3 in Appendix A). This confirms that noise rate is also the dominant factor in governing the algorithm correctness for Grover circuits specifically, in addition to its influence on the output quality (TVD).

9.7 Runtime Distribution by Circuit Type

While runtime was included as a node in the causal graph (Section 8), it was not a target of the arrow strength or counterfactual analyses presented above, which focused on TVD and success probability as the primary outcome metrics. As an observation, runtime distributions varied modestly across circuit types: median runtime ranged from approximately 0.003 seconds for Bell, GHZ, and Parameterized circuits to approximately 0.005 seconds for Grover and Variable Depth circuits, with all five circuits showing a similar overall shape: most runs completed quickly, with a small number of slower outlier runs extending up to roughly 0.04 seconds in most circuits (Figure 12 in Appendix A). This is consistent with Grover and Variable Depth being the deepest and most gate-dense circuits in the benchmark suite, though a formal causal analysis of runtime was not conducted in this project.

9.8 Circuit-Specific Sensitivity to Noise Rate

While noise rate degrades output quality across all five benchmark circuits, the magnitude of degradation varies substantially by circuit structure. Holding circuit type constant, mean TVD at noise_rate = 0.05 ranged from 0.059 for the Bell state to 0.436 for Grover Search. This is around a 7-fold difference despite both circuits experiencing the same noise conditions (Figure 10). Grover and Variable Depth, the two circuits with the greatest entanglement depth and two-qubit gate density, showed the sharpest degradation, consistent with noise accumulating over longer gate sequences.

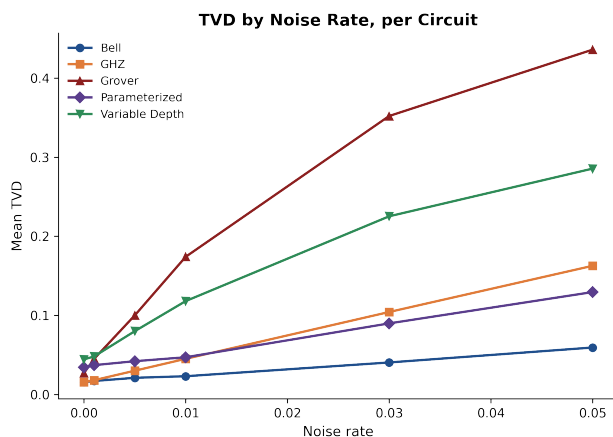


Figure 10: TVD as a function of noise rate, broken out by circuit type. Grover and Variable Depth show the steepest degradation.

10 Prototype Tool

A Streamlit dashboard was developed to provide an interactive interface for exploring the experiment dataset, causal graph, and counterfactual findings described in earlier sections. The dashboard is implemented in Python (see Code Repository in Appendix B), reusing the benchmark circuit code from Task 2, the causal graph from Task 5, and the fitted DoWhy causal models from Task 6. It is organized into five tabs: Data Explorer, Causal Graph, Counterfactuals, Circuit Diagrams, and About.

The dashboard supports the functions specified for this task as follows:

- **Selecting circuits and choosing configurations:** a sidebar lets the user select a benchmark circuit and configure the causal variables (qubit count, shots, noise rate, and optimization level) via sliders. These are visually separated from a second sidebar section, “Other Configurable Parameters,” which exposes circuit-specific but non-causal settings: circuit depth for Variable Depth, rotation axis/angle per qubit for Parameterized, and target state for Grover, with captions clarifying that these affect only the live circuit diagram, not the filtered dataset.
- **Running simulations and storing execution data:** live Qiskit simulation was not included in this dashboard in favor of exploring the existing 12,672-row dataset generated in Tasks 3 – 4; this decision is discussed further in the Section 13.
- **Visualizing metrics and relationships:** the Data Explorer tab (Figure 14 in Appendix A) displays the dataset rows matching the selected configuration, summary statistics (mean, standard deviation, min, max) for each outcome metric, and a TVD distribution chart.
- **Displaying a causal graph:** the Causal Graph tab displays the causal graph from Section 8, with a brief caption explaining the node and edge semantics.
- **Comparing estimated effects:** the Counterfactuals tab (Figures 15 and 16 in Appendix A) lets the user select an intervention variable (noise rate or optimization level) and a new value, then runs a live counterfactual query against the fitted causal models and displays the observed and counterfactual outcomes side by side, along with the percentage change.

As an additional feature beyond the required functions, a Circuit Diagram tab (Figure 13 in Appendix A) dynamically renders the actual benchmark circuit for the selected configuration.

11 Results

This section summarizes the main findings from the experiments and analysis conducted for this project, as described in Sections 6 through 9.

Dataset scale. The experiment pipeline generated 12,672 executions, spanning five benchmark circuits (Bell, GHZ, Parameterized, Grover, and Variable Depth), with each circuit varying the following workflow configurations: qubit count (2 – 5), shots (128 – 4096), noise rate (0.0 – 0.05), and optimization level (0 – 3), with circuit-specific parameters (depth, target state) varied where applicable (Table 2).

Metric summary. Averaged across all circuits and configurations, the mean TVD was 0.10445. TVD ranged from 0.0155 (Bell state circuit, zero noise) to 0.4359 (Grover circuit, noise_rate = 0.05). See Figure 17 in Appendix A.

Strongest configuration effects. Ranking the causal variables by their estimated arrow strength on TVD (using DoWhy) produced the following ordering: `noise_rate` (0.006524), `depth` (0.002445), `count_2q` (0.001168), `num_qubits` (0.000329), and `shots` (0.000230), shown in Figure 9.1. Shots had the weakest effect among the five variables (see Figure 11 in Appendix A). The `optimization_level` showed no meaningful direct effect on TVD, a result that held across the counterfactual analysis as well (Section 9).

Key causal findings. Two counterfactual interventions produced the most substantial results:

- Fixing `noise_rate` to 0.001 reduced mean TVD (averaged across all five circuits) from 0.10445 to 0.03412, a reduction of 67.3%, the largest effect observed among all tested interventions (Figure 9).
- Under the same low-noise condition, the Grover circuit’s success probability changed from 0.57483 to 0.73580, an improvement of approximately 28%.

By contrast, fixing `optimization_level` to its maximum value of 3 reduced mean TVD by less than 1%.

Circuit-specific variation. The magnitude of TVD change with noise rate varied by circuit type. At `noise_rate` = 0.05, mean TVD ranged from 0.0593 for the Bell state circuit to 0.4359 for the Grover circuit (Figure 17 in Appendix A), which is roughly a 7-fold difference despite identical noise conditions (Figure 10). Grover and Variable Depth, the circuits with the greatest entanglement depth and two-qubit gate density, showed the largest TVD values at high noise rate.

Table 3 in Appendix A summarizes the three headline counterfactual results discussed above.

12 Discussion

The results presented in Section 11 point to a clear story: among the five workflow configurations studied, noise rate is the most consequential choice for quantum program output metrics, especially output quality and success probability. Qiskit’s transpilation optimization level, despite being the configuration most directly associated with “quality of code” in classical programming intuition, has almost no measurable effect at the benchmark circuit tested here.

Why noise rate is the strongest. This finding does make physical sense: noise rate directly perturbs qubit state during every gate operation in a circuit, whereas `depth` and `count_2q` only create the *opportunity* for that noise to accumulate. In other words, noise rate is the root cause of output degradation, while `depth` and `count_2q` act as amplifiers of that same underlying noise. This distinction is reflected directly in the causal graph structure (Section 8), where `noise_rate` has a direct edge into TVD, while `depth` and `count_2q` sit as mediating variables between the root causal variables and the outcome metrics.

Why optimization level has negligible effect. At first glance, the counterfactual analysis showing around 1% reduction in TVD from the maximum optimization level was counterintuitive. For comparison, in classical computers, more aggressive optimization typically improves performance meaningfully. However, the scale of benchmark circuits studied for this project is the main explanation: with only 2 to 5 qubits and depth up to 15, there is a limited redundancy for Qiskit’s transpiler to exploit. Gate cancellation and circuit restructuring, which are the primary actions done by Qiskit’s transpilation optimizer, have more to work with in larger, more redundant, and complex quantum circuits. Optimization level simply does not have enough structural room to simplify or combine gates sufficiently to have a significant effect on TVD to compete with noise rate’s direct influence. This suggests that optimization level’s importance may be more prominent on more complicated circuits and is substantially understated by this study (see Section 13 and 14).

Consistency and variation across circuits. While noise rate’s effect was consistent across all five circuits, where higher noise caused higher TVD, the magnitude of that effect varied substantially. For example, the TVD between the Bell state circuit and the Grover circuit showed a 7-fold difference at high noise rate. This variation is not surprising given the circuit’s structural differences: Grover involves a deeper and more entanglement-heavy circuit structure with higher two-qubit gate density than Bell’s single CNOT. Since two-qubit gates carry higher error rates than single-qubit gates in the depolarizing noise model used here, circuits that lean more heavily on entanglement are inherently more exposed to accumulated noise. This is consistent with the causal graph, in which `count_2q` mediates a meaningful portion of noise’s effect on TVD, and explains why circuit identity itself (`circuit_name`) was chosen as a causal variable.

Noise rate and algorithmic correctness. The Grover-specific counterfactual result extends the noise-rate findings beyond TVD (output quality) to a second, more applicable metric: whether Grover’s algorithmic circuit actually identifies the correct marked state. A 28% improvement in success probability under low noise demonstrates that noise rate has an impact that extends to metrics other than TVD. Therefore, this result argues strongly for prioritizing noise reduction over transpilation tuning, at least for circuits that are comparable to the ones studied here.

Benefit of the causal graph as an organizing framework. Beyond the specific numerical findings, the causal graph (see Section 6) proved to be helpful as a structuring model for the analysis in a way that a purely correlational approach would not have. By modeling depth and `count_2q` as mediators rather than independent causes, the graph made it possible to distinguish between a variable’s total downstream influence and its practical significance. For example, the causal graph visually shows how the optimization level has an indirect influence over the outcome metrics, rather than a direct influence, especially on the TVD. Without this structure, it would have been easy to either overstate that the optimization level is important based on its correlation with depth and 2-qubit gate count mediators, or to miss the distinction between the noise rate’s role as a root cause versus depth and 2-qubit gate count’s role as amplifying mediators. Additionally, the causal graph made the counterfactual queries in Section 9 possible in the first place, since counterfactual reasoning requires an explicit causal structure to reason over, unlike simple conditional comparisons or correlations. This directly addresses whether a simple causal graph can help explain how quantum workflow configurations influence observed outcomes.

Collectively, these results suggest that for small-to-moderate-scale quantum circuits, workflow-level effort is best spent reducing noise exposure rather than increasing Qiskit’s optimization level, and that causal modeling, even at the relatively simple level applied to this project, can meaningfully clarify which configuration choices actually matter, as opposed to which ones merely correlate with better outcomes.

13 Limitations

Several limitations should be considered when interpreting the results of this project:

Small circuit scale. The benchmark suite was intentionally limited to small circuits (2 – 5 qubits, depth up to 15) to keep simulation and analysis reasonable within a nine-week program. As noted in Section 9, this likely explains why optimization level showed a negligible effect on output quality; larger, more complex circuits would give the transpiler more opportunity to meaningfully reduce gate count and depth, which may in turn produce a stronger downstream effect on TVD (output quality).

Simulator-based experiments. All experiments were run on Qiskit’s AerSimulator, rather than physical quantum hardware (like IBM Quantum Backend). While the depolarizing noise model used here approxi-

mates real device behavior, it does not capture the full complexity of hardware-specific noise sources such as crosstalk, readout error correlations, or device-specific gate calibration drift.

Simplified noise model. The project used a single depolarizing noise model with a scalar noise rate parameter. Real quantum devices exhibit more complex, heterogeneous noise structures that vary by qubit, gate type, and time. A more realistic noise model derived from actual backend calibration data was outside the scope of this project.

Limited configuration space. Other factors known to affect quantum program behavior, such as backend topology, qubit connectivity constraints, and alternative transpilation passes, were not included in the causal graph, in addition to the configurations that were studied.

Simple causal methods. The causal analysis relied on DoWhy’s graphical causal model framework with gradient-boosted regression mechanisms and arrow strength estimation. While effective for the scope of this project, more sophisticated causal discovery or estimation methods (e.g., automated structure learning) were not explored, and the causal graph structure itself was specified manually.

No hardware validation. Due to time constraints, no experiments were validated against real IBM Quantum Backends. Therefore, all findings are specific to the simulator and noise model used, and their generalization to physical devices remains untested.

No live simulation in the dashboard. The prototype tool (Section 10) exposes configuration controls for running Qiskit simulations, but does not execute new simulations live. It explores the pre-generated dataset from Section 6 instead. Implementing live simulation would require modularizing the noise model construction, transpilation, execution, and metric computation logic currently embedded in the Task 3 – 4 notebooks into reusable functions, which was deferred given the project’s timeline. This means the dashboard cannot currently answer queries for configuration combinations outside the 12,672 runs already collected.

Short project timeline. The nine-week program timeline constrained the scope of the benchmark suite, configuration space, and tool development. Several extensions that would have strengthened the project’s conclusions, discussed in Section 14, were identified but not completed within this timeframe.

14 Future Work

Several extensions could build on the current project and address the limitations described in Section 13:

Real IBM Quantum Backend experiments. Validating the current findings on physical quantum hardware would confirm whether the causal relationships identified here, particularly the dominance of noise rate over optimization level and shot count, hold under real device conditions, or whether hardware-specific effects change the picture by introducing nuance.

Backend-derived noise models. Rather than a single scalar depolarizing noise rate, future work could construct noise models directly from IBM Quantum backend calibration data, capturing per-qubit and per-gate error rates, readout error, and crosstalk. This would allow the causal graph to be re-evaluated under a more realistic and heterogeneous noise structure.

Larger circuits and additional benchmarks. Extending the benchmark suite to larger qubit counts and deeper circuits would test whether the optimization level’s effect on output quality becomes more pro-

nounced as circuits grow in complexity, as hypothesized in Section 13. Additional benchmark circuits, such as quantum Fourier transform, quantum teleportation, or variational quantum algorithms, would also broaden the generality of the causal findings beyond the five circuits studied here.

Improved causal effect estimation. Future work could explore automated causal structure learning to validate or refine the manually specified causal graph, as well as more sophisticated causal mechanism models beyond gradient-boosted regression, potentially improving the precision of arrow strength and counterfactual estimates.

Automated experiment planning. Rather than a fixed configuration grid, an automated experiment planner could adaptively select which configurations to run next based on current causal uncertainty, focusing simulation budget on the most informative regions of the configuration space.

Better tool support. A great next step from the Streamlit tool (see Section 10) is extending the dashboard to support running new simulations directly from the interface, building on the benchmark suite developed in this project, along with richer causal graph editing and comparison features.

15 Conclusion

This project was designed to study how various workflow configurations in quantum circuits causally influence program outcome metrics such as TVD, runtime, and success probability, using controlled experiments and causal analysis without developing a new quantum algorithm. A small benchmark suite of five circuits was implemented, a dataset of 12,672 executions was collected across systematically varied configurations, and a causal graph connecting these configurations, mediating variables, and outcome metrics was constructed and analyzed using DoWhy’s graphical causal model framework.

The analysis showed that the simulated noise rate is the most consequential configuration for output quality, with a direct counterfactual intervention reducing the mean TVD by 67.3% and improving the Grover search success probability by approximately 28%. Transpilation optimization level and shot count had comparatively weak effects at the circuit scale studied here. Additionally, different circuit types reacted differently to noise. Runtime was measured and included in the causal graph, but was examined only descriptively. Beyond these specific findings, this project demonstrates the value of causal and counterfactual analysis for understanding quantum computing workflows. A purely correlational analysis of the same dataset could have shown that optimization level correlates with reduced depth and gate count, but would not have distinguished this correlation from a meaningful downstream effect on output quality, nor would it have supported the counterfactual “what if” queries that gave this project’s findings their practical interpretability. By explicitly modeling the quantum workflow using a causal graph, this project identified what configurations matter, as well as why and by how much. This provided a foundation for prioritizing noise mitigation.

16 References

- [1] IBM Quantum Learning, “Learn quantum computing with Qiskit.” [Online]. Available at: <https://quantum.cloud.ibm.com/learning>
- [2] IBM Quantum Documentation, “Qiskit documentation.” [Online]. Available at: <https://www.ibm.com/quantum/qiskit>
- [3] Qiskit Aer Documentation, “Building noise models.” [Online]. Available at: https://qiskit.github.io/qiskit-aer/tutorials/3_building_noise_models.html

- [4] Qiskit Aer Documentation, "AerSimulator." [Online]. Available at: https://qiskit.github.io/qiskit-aer/stubs/qiskit_aer.AerSimulator.html
- [5] K. Sharma, J. Chappell, C. Blank, et al., "DoWhy: An End-to-End Library for Causal Inference." [Online]. Available at: <https://github.com/py-why/dowhy>
- [6] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*. Cambridge, UK: Cambridge University Press, 2010.

A Appendix

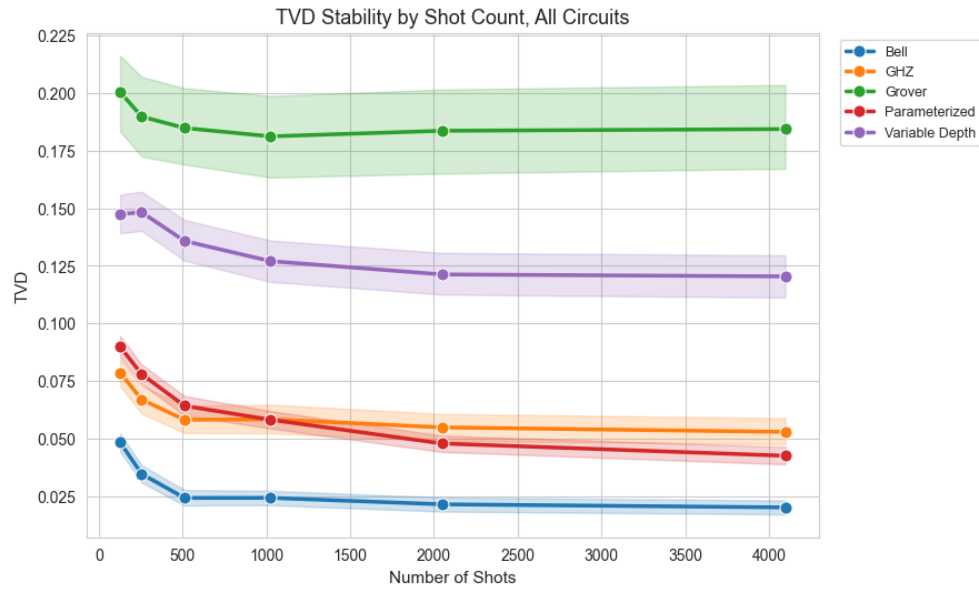


Figure 11: TVD across varying shot counts, illustrating the comparatively weak effect of shot count on output quality relative to noise rate and circuit depth.

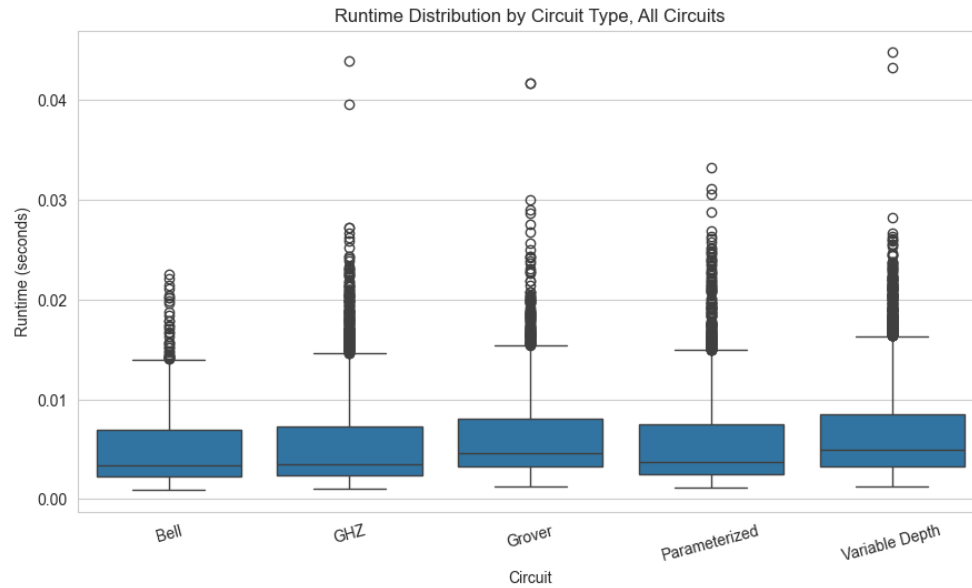


Figure 12: Runtime distribution by circuit type across all runs, showing median, interquartile range, and outliers for each of the five benchmark circuits.

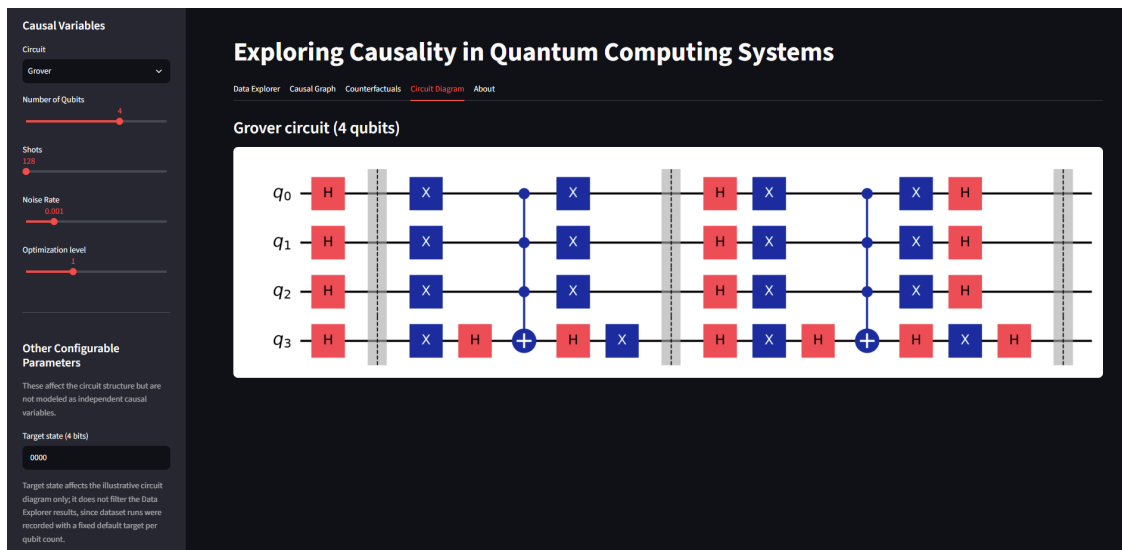


Figure 13: Circuit Diagram tab, rendering a live-built Grover circuit with a user-specified 4-qubit configuration, alongside the sidebar’s non-causal configuration controls.

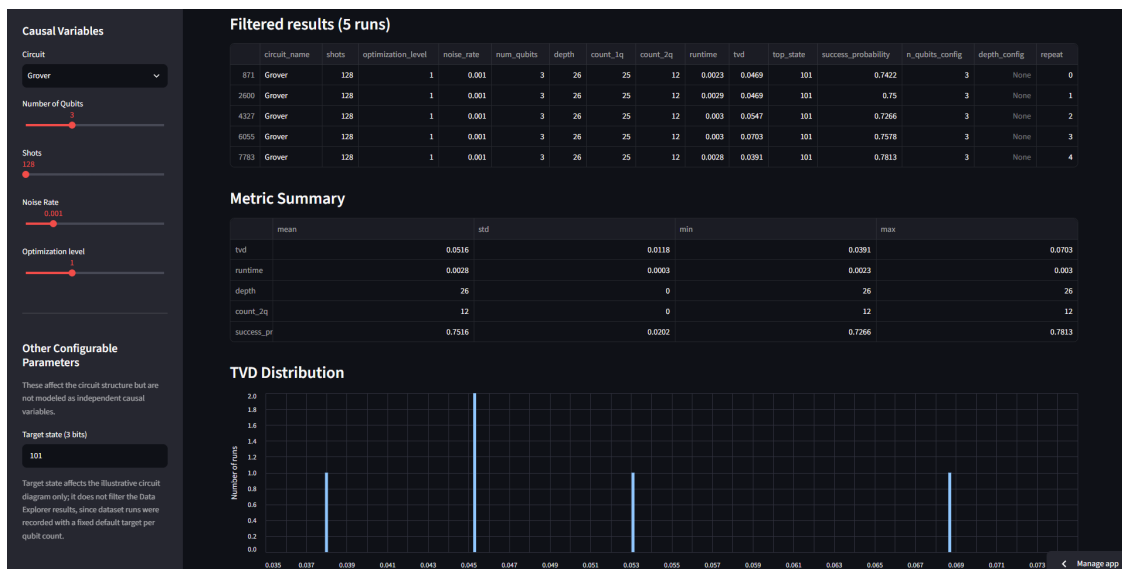


Figure 14: Data Explorer tab, showing filtered dataset results and computed metric summaries for a selected Grover configuration.



Figure 15: Counterfactuals tab, showing a live counterfactual query comparing observed and counterfactual mean TVD and success probability under a fixed noise rate intervention.



Figure 16: Counterfactuals tab, showing a live counterfactual query comparing observed and counterfactual mean TVD under a fixed optimization level.

Table 3: Counterfactual Query Results

Outcome	Condition	Observed	Counterfactual	Change
TVD (avg., all circuits)	noise_rate = 0.001	0.10445	0.03412	67.3%
TVD (avg., all circuits)	optimization_level = 3	0.10445	0.10331	~1%
Success probability (Grover)	noise_rate = 0.001	0.57483	0.73580	28%

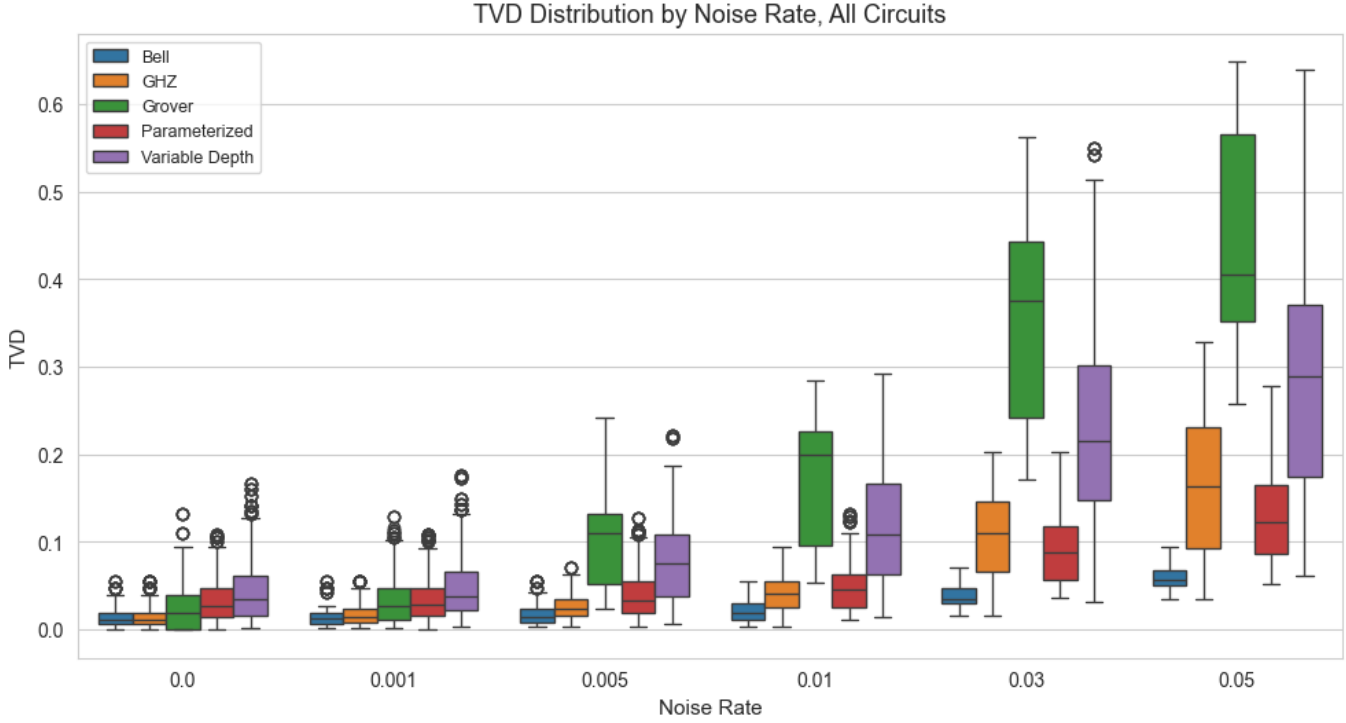


Figure 17: TVD varied depending on circuit type and noise rate.

B Code Repository

The complete source code, benchmark circuits, and replication scripts for this project are publicly available on GitHub at:

https://github.com/VigneshT24/AISys_QC_Exploring_Causality